

Task 1: Genomic Variants

COURSE NAME

CSDS 413 Introduction to Data Analysis

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Context

In this exercise, our aim is to quantify the association between genomic variants in the plant *Arabidopsis Thaliana* while assessing the statistical significance of their association. In this context, the variables are genomic variants and the samples are individual plants. We will use the two datasets that are provided with the assignment:

- The file "p1a.csv" contains a binary matrix of size 199 x 2 which indicates the presence of two genomic variants for 199 individuals. A value of 1 indicates that the genomic variant is present for the corresponding individual, and 0 indicates it is not.
- The file "p1b.csv" contains a binary matrix of size 199 x 15 which has 199 samples (individuals) and 15 variables (genomic variants). Each column represents a variable, and each entry (1 or 0) in the column indicates the presence of the corresponding variable in each of the 199 samples.

1 Part (a)

For the two variables provided in "p1a.csv", assess the association between them by computing each of the following statistics:

- Mutual Information
- Jaccard Index
- Pearson's chi-squared χ^2

For each of these statistics, assess the statistical significance by computing a p-value for the null hypothesis of "there is no association" (see *Note (1)* below). Select a significance level α that we can use to reject the null hypothesis and accept the alternate hypothesis "there is a non-zero association." if a p-value is less than α (see *Note (2)* below for the selection of α). For each of the three specified statistics, explain, in complete sentences, your findings: Is there a statistically significant association between the given variables? How strong does the association seem? Do the results of different statistics agree? If not, what does this mean (i.e., how do you explain this discrepancy)? Make sure to specify the values of the test statistics and the p-values as well as the selected α level and the number of permutations N you use for the permutation tests.

Note (1): The p-value in this context indicates the probability of observing a "more significant" result (for a statistic of interest like mutual information) than the observed result when there is no association between the genomic variants (which is our null hypothesis).

Note (2): You are welcome to select the commonly used 0.05 as your α threshold. However, you are encouraged to try different α values and observe how the conclusions of a researcher might change dramatically if they are not careful with the interpretations of the hypothesis test results.

Hint (1): For mutual information and Jaccard index, you need to do a Monte Carlo simulation (e.g., a permutation test) to assess statistical significance. For Pearson's χ^2 , you can use the parametric distribution (i.e., [Pearson's chi-squared test](#)) instead of doing a Monte-Carlo simulation to generate the null distribution.

Hint (2): The idea of the permutation test is to randomly permute the entries in each column (variable) separately, in order to break any possible associations between the variables without modifying their distributions. This way, we can generate a null distribution for the test statistic (like mutual information) that represents our null hypothesis. Performing the permutation test is simple: First, generate N random permutations of the data and for each of these permuted data, compute the test statistic of interest. Then, count the number of times permuted data has more significant values (e.g., higher values for mutual information) than the actual data, let's denote this count c . Then, the p-value is simply equal to $\frac{c+1}{N+1}$. Notice that, a p-value obtained from a permutation test cannot be lower than $\frac{1}{N+1}$.

Hint (3): An alternative to a permutation test (or Monte-Carlo simulation in general) is to obtain a null distribution analytically by fitting a known parametric distribution. For example, it is known that Pearson's χ^2 follows a well-studied distribution called [chi-squared distribution](#) when the null hypothesis of "no association" is correct. Thus, it is possible to obtain a p-value analytically from the chi-squared distribution (indeed, this is what the [Pearson's chi-squared test](#) does).

Hint (4): To visualize the results of a permutation test, you can draw a sorted scatter plot while marking the observed value (as shown in 'example-figure1.jpg').

To perform the permutation test framework on the p1a dataset for mutual information and Jaccard index, I set $N = 10,000$. For Pearson's χ^2 , I decided to follow the suggestion in the instruction and analytically fit the parametric chi-squared distribution with 1 degree of freedom.

For all three tests, I used $\alpha = 0.05$ to reject the null hypothesis, below is the implementation of the three statistics and the relevant tests:

```
def mutual_information(x: np.ndarray, y: np.ndarray) -> float:
    """
    Calculates mutual information between two binary variables.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
```

```

:type y: np.ndarray
:returns: Mutual information value
:rtype: float
"""
contingency = contingency_table(x, y)
n = len(x)
p_joint = contingency / n

p_x = p_joint.sum(axis=1)
p_y = p_joint.sum(axis=0)

mi = 0
for i in range(2):
    for j in range(2):
        if p_joint[i, j] > 0:
            mi += p_joint[i, j] * np.log2(p_joint[i, j] / (p_x[i] * p_y[j]))

return mi

def jaccard_index(x: np.ndarray, y: np.ndarray) -> float:
    """
    Calculates Jaccard Index between two binary variables.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: Jaccard Index value
    :rtype: float
    """
    intersection = np.sum((x == 1) & (y == 1))
    union = np.sum((x == 1) | (y == 1))

    if union == 0: return 0

    return intersection / union

def chi_squared_statistic(x: np.ndarray, y: np.ndarray) -> float:
    """
    Calculates Pearson's chi-squared statistic.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: Chi-squared statistic
    :rtype: float
    """
    contingency = contingency_table(x, y)
    chi2_stat, _, _, _ = chi2_contingency(contingency)
    return chi2_stat

def contingency_table(x: np.ndarray, y: np.ndarray) -> np.ndarray:
    """
    Creates 2x2 contingency table for two binary variables.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: 2x2 contingency table

```

```

:rtype: np.ndarray
"""
table = np.zeros((2, 2))
table[0, 0] = np.sum((x == 0) & (y == 0))
table[0, 1] = np.sum((x == 0) & (y == 1))
table[1, 0] = np.sum((x == 1) & (y == 0))
table[1, 1] = np.sum((x == 1) & (y == 1))
return table

def permutation_test(x: np.ndarray, y: np.ndarray, statistic_func,
                    n_permutations: int) -> tuple:
    """
    Performs permutation test to calculate p-value for given statistic.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :param statistic_func: Function to calculate test statistic
    :type statistic_func: function
    :param n_permutations: Number of permutations
    :type n_permutations: int
    :returns: Tuple of observed statistic, p-value, null distribution
    :rtype: tuple
    """
    observed_stat = statistic_func(x, y)

    null_dist = np.zeros(n_permutations)

    for i in range(n_permutations):
        y_permuted = np.random.permutation(y)
        null_dist[i] = statistic_func(x, y_permuted)

    c = np.sum(null_dist >= observed_stat)
    p_value = (c + 1) / (n_permutations + 1)

    return observed_stat, p_value, null_dist

def chi_squared_parametric(x: np.ndarray, y: np.ndarray) -> tuple:
    """
    Performs Pearson's chi-squared test using parametric distribution.

    :param x: First binary variable
    :type x: np.ndarray
    :param y: Second binary variable
    :type y: np.ndarray
    :returns: Tuple of chi-squared statistic, p-value
    :rtype: tuple
    """
    contingency = contingency_table(x, y)
    chi2_stat, p_value, _, _ = chi2_contingency(contingency)

    return chi2_stat, p_value

```

Figure 1: Core functions for computing statistics and performing permutation tests.

The computed test statistics and corresponding p-values:

Statistic	Value	P-value
Mutual Information	0.047	0.0027
Jaccard Index	0.000	1.0000
Pearson's χ^2	6.456	0.0111

Figure 2: Test statistics and p-values for the two genomic variants in p1a.csv.

I didn't see a example-figure1.jpg anywhere for reference, but a sorted scatter plot sounds straightforward enough for MI and JC, below are the plots for each, with each of the observed statistics marked with a dashed red line.

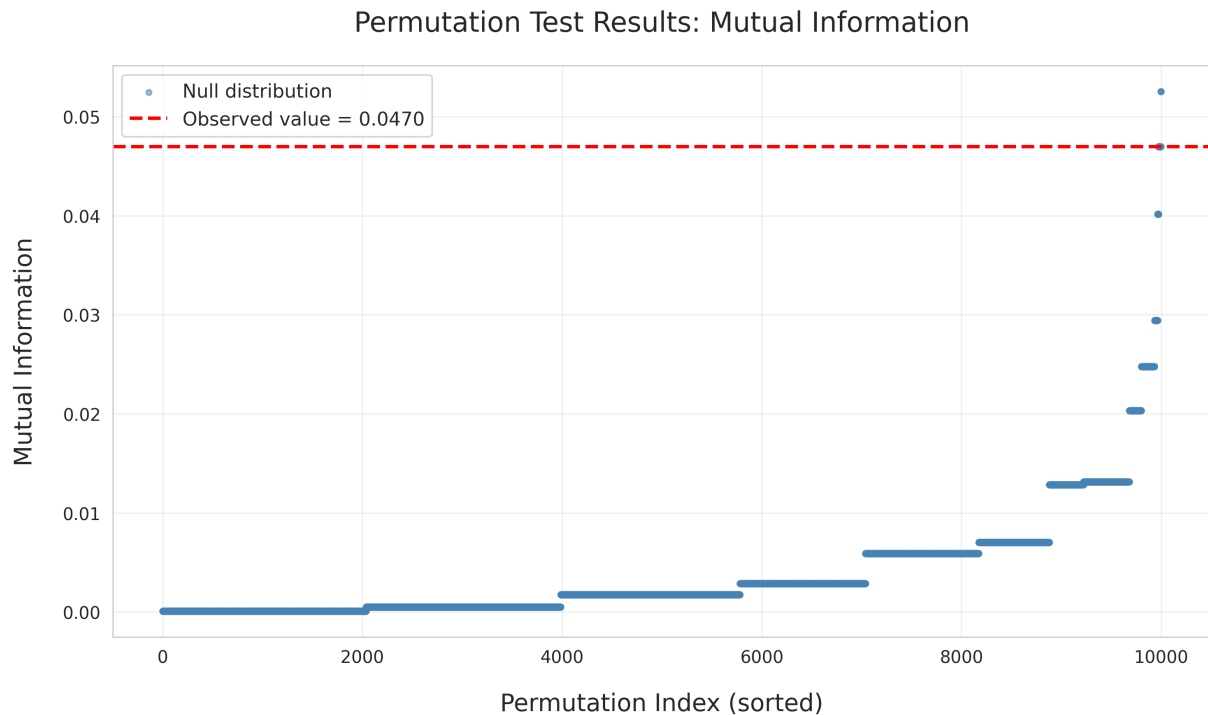


Figure 3: Permutation test results for Mutual Information.

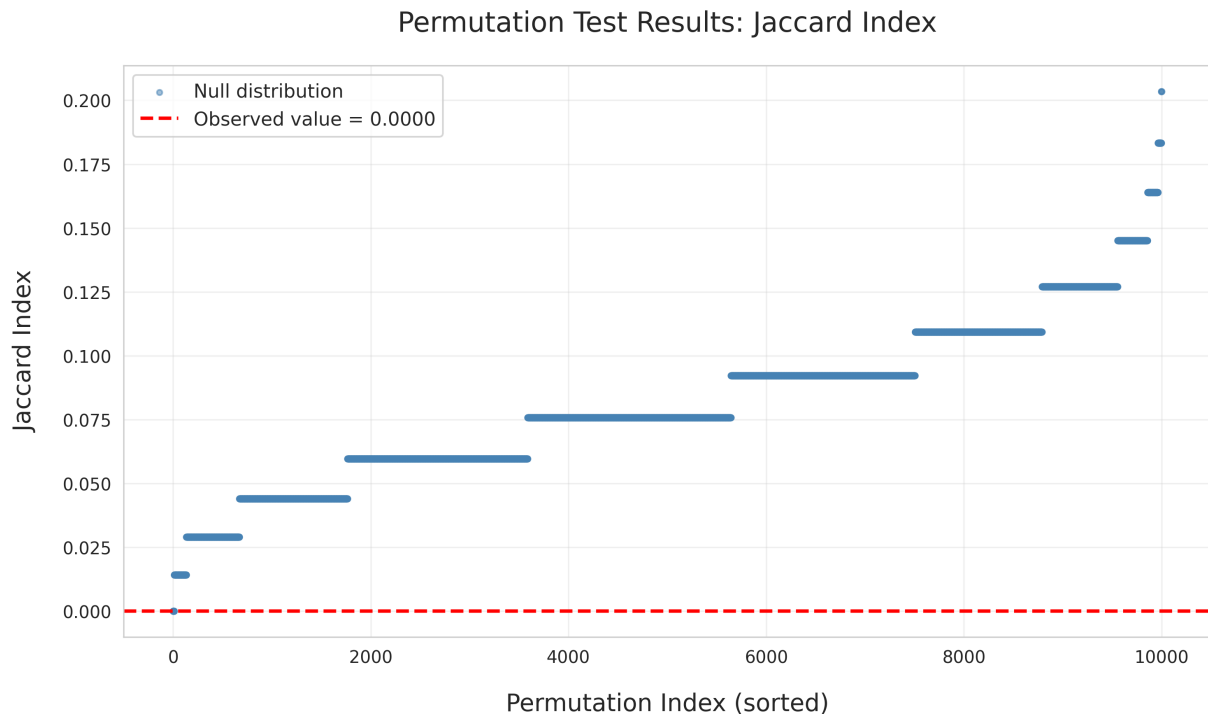


Figure 4: Permutation test results for Jaccard Index.

The MI of 0.047 is small, indicating that there is a small, but statistically detectable ($p = 0.0027$), amount of information shared between the variables; they are just a bit telling of each other. The chi-squared also shows a weak yet significant ($p = 0.0111$) association between the two variables with a value of 6.456, varying from what we ought to expect under independence. The Jaccard on the other hand, shows no association and upon inspection of the data following this result, I realized there is no occurrence of (1,1) in the data, yielding a value

of 0 and a p value of 1.

This means the statistics do not all agree, but there is good reason for this; it's a demonstration of the fact that these statistics capture different aspects of association between two variables, and in this specific case, we have a set of data that doesn't align well necessarily with how these statistics are defined. Jaccard is specifically helpful in the case where there is a relation between the (1,1) class and the union in the contingency table, but in this example, it is likely that the detection of any association is a result of the samples being mutually exclusive rather than the variables ever both being 1 at the same time. For this reason, MI and χ^2 are equipped to identify the occurrence of one variable given the result of another and the difference between the expected and observed values under independence, while the Jaccard is left blind to the connection by its construction.

2 Part (b)

For each of the 105 ($15 \times 14/2$) pairs of variables provided in "p1b.csv", repeat the steps in part (a) to compute the values of the test statistics as well as the p-values separately for: Mutual information, Jaccard Index and Pearson's chi-squared χ^2 statistics. Select an α level to reject the null hypothesis of no association. For each of the three test statistics, determine the pairs with significant association at α level while adjusting for false discovery rate (see Note (3) below). Draw scatter plots (or other appropriate visualizations) to compare the results of mutual information (MI), Jaccard Index (JI) and Pearson's chi-squared χ^2 . How much do the results of these three test statistics agree to one another? Which two of these three statistics are most similar (MI-JI, MI- χ^2 , or JI- χ^2)? If you were to use only one of these three statistics, which one would you prefer to use and why? If you were to consider only your preferred test statistic among the given three, would your conclusions (about the associations of the provided variable pairs) change a lot? Make sure to specify the number of significantly associated pairs and the number of overlap between the three test statistics. Also, specify the selected α level and the number of permutations N you use for the permutation tests.

Note (3): Here, differently from part (a), we are testing for multiple hypotheses (one for each variable pair). Therefore, if we were to reject a null hypothesis as suggested in part (a) (when p-value is less than α), we would have a considerably higher **false discovery rate (FDR)** than α (Think: If each test has a α chance of failure, what is the probability that there is at least one test that fails? For $\alpha = 0.05$ and $n = 105$ tests, that would be 99.54% chance of failure!). Thus, we need to use an appropriate procedure that take into account the number of hypotheses to make sure the FDR is indeed less than α . For this purpose, you can apply one of the following two approaches: (1) **Benjamini-Hochberg** procedure (which is, in many fields, the go-to way of adjusting for multiple hypotheses) to determine the statistically significant pairs, (2) A Monte Carlo simulation specifically designed to control the FDR based on the distributions generated by the permutations (e.g., to compute the p-value of the top-scoring pair, we can use the distribution of the test statistics of the top-scoring pairs across all permutations, which would be a "multiple-hypothesis-corrected" null distribution).

Note (4): Using the Benjamini-Hochberg procedure, essentially a stricter threshold is used to reject a null hypothesis (e.g., the threshold of $\frac{\alpha}{n}$ for rejecting the null hypothesis of the most significant pair). Therefore, for estimating the p-values using permutation test, you need to use more permutations compared to part (a) to be able find a significant association. Thus, make sure that your number of permutations N is high enough to be able to find at least one significant pair.

For the purpose of testing for multiple hypotheses and controlling for FDR, I applied the Benjamini-Hochberg (BH) procedure at $\alpha = 0.05$: to identify the most significant associations after FDR correction, we need enough permutations to reliably estimate p-values at about $0.05/105 \approx 0.0005$; towards this end, I set $N = 50,000$. Figure 5 gives the number of significant pairs detected by each statistic and Figure 6 gives the overlap between them:

Statistic	Significant Pairs
Mutual Information	91
Jaccard Index	56
Pearson's χ^2	90

Figure 5: Number of significant pairs detected by each statistic after Benjamini-Hochberg FDR correction at $\alpha = 0.05$.

Comparison	Number of Overlapping Pairs
MI $\cap$ JI	55
MI $\cap$ χ^2	89
JI $\cap$ χ^2	56
MI $\cap$ JI $\cap$ χ^2	55

Figure 6: Overlap of significant pairs between different statistics.

Below are the scatter plots side-by-side comparing each of the statistics:

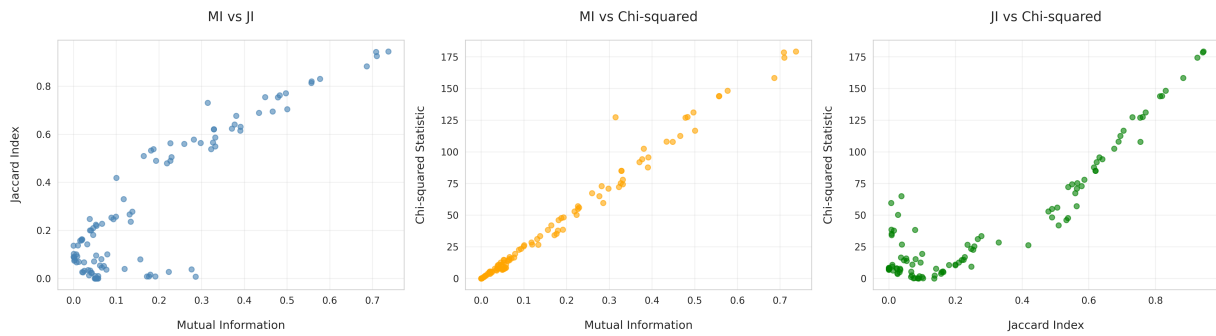


Figure 7: Pairwise comparison of test statistics across 105 genomic variant pairs.

To characterize each pair from the visual: MI and chi-squared are very similar, showing a very high level of agreement. Jaccard relative to the other two show similar and moderate levels of agreement, but both show some level of difference at lower extremes.

Unsurprisingly from the discussion in the previous part, MI and chi-squared show a high degree of agreement while each agrees somewhat with Jaccard. Again, because both of these statistics use all four cells of the contingency table to represent deviation from independence, and Jaccard is constructed more as a similarity measure that is blind to the mutual exclusivity pattern in the data, their ability to identify associations in this data are both better than Jaccard and most similar to each other.

I would say that using either MI or Pearson's chi-squared is going to generally arrive at the same result; the only case where just using one statistic would get you very different conclusion would be Jaccard. However, the computational efficiency of using chi-squared and not having to do many permutation tests to get essentially the same result would make it my preferred statistic. I think the difference in significant pairs by one may be more telling with additional exercises, but its not compelling enough given this test to believe that MI is noticeably superior.